Hospital Billing and Patient Classification

Complete worked solution

Sample policy: bill consultation, diagnostics, and treatment; apply a 10% concession to the whole bill for age 60+ or verified low-income patients (one concession only). Validate nonnegative quantities and rates, age, and payment between zero and final bill. Classify as PAID, PARTLY PAID, or UNPAID; any balance above 50000 is additionally HIGH BALANCE.

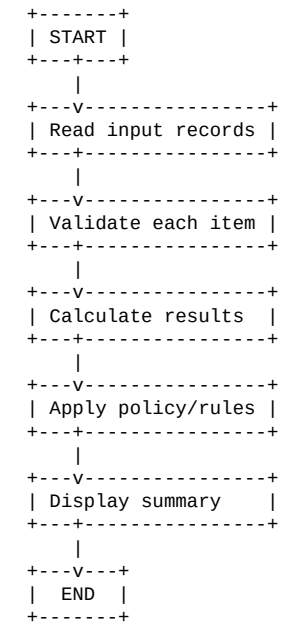
1. Computational Thinking and Algorithm

Inputs: Supplied records and policy values. **Processing:** Validate data, calculate charges or scores, apply ordered rules, and compare outcomes. **Outputs:** Itemized results, classifications, rankings, or recommendations.

Pseudocode

```
START
  Read patient details and service quantities/rates
  Validate inputs and payment constraints
  Calculate consultation + diagnostic + treatment charges
  IF age >= 60 OR low-income status is true, apply 10% concession
  Calculate amount due and balance
  Classify payment and balance using policy
  Print itemized bill and classification
END
```

2. Flowchart



3. Python Program

The program uses basic variables, numeric and string data, tuples, lists, loops, conditionals, arithmetic, and formatted output. Policy values are constants near the top of the program.

```
patients = [
    ("P201", "Asha", 67, True, 2, 750, 1, 2400, 3, 1800, 7000),
    ("P202", "Ben", 34, False, 1, 900, 2, 1200, 1, 3500, 2000),
]
# record: id, name, age, low_income, consult_qty, consult_rate,
#         test_qty, test_rate, treatment_qty, treatment_rate, paid
for record in patients:
```

```
(pid, name, age, low_income, cq, cr, dq, dr, tq, tr, paid) = record
values = [age, cq, cr, dq, dr, tq, tr, paid]
valid = True
for value in values:
    if value < 0:
        valid = False
if age > 120 or not valid:
    print(pid, "invalid patient or charge data")
    continue
consultation = cq * cr
diagnostics = dq * dr
treatment = tq * tr
gross = consultation + diagnostics + treatment
concession = gross * 0.10 if age >= 60 or low_income else 0
due = gross - concession
if paid > due:
    print(pid, "payment exceeds amount due; record rejected")
    continue
balance = due - paid
if balance == 0:
    status = "PAID"
elif paid > 0:
    status = "PARTLY PAID"
else:
    status = "UNPAID"
if balance > 50000:
    status = status + " / HIGH BALANCE"
print("Patient:", name, "ID:", pid)
print("Consultation:", consultation, "Diagnostics:", diagnostics,
      "Treatment:", treatment)
print("Gross:", gross, "Concession:", concession,
      "Due:", due, "Paid:", paid, "Balance:", balance)
print("Classification:", status)
```

4. Sample run guidance

Run the program as written to see a sample result. Replace the sample lists and constants with the values provided in your assignment. Keep the rule order and units consistent; for example, percentages should be entered as percentages and converted to fractions only in the calculation.